Spatiotemporal Variation of Imaging Swath in Earth Observing Lunar-Based SAR Under Orbital Perturbations

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***Abstract*— We analyzed the spatiotemporal variations of the imaging swath and region using the lunar-based synthetic aperture radar (LBSAR) for Earth observation. The ground and slant swaths following the Earth-observing geometry and synthetic aperture radar (SAR) configurations were formulated. The bounds of the imaging swath were highlighted for appropriately configuring the LBSAR. The results suggested that the imaging swath is determined by the coupling of orbit elements, near-look angle, antenna beamwidth, and to some extent by the Earth's ellipsoidal shape and LBSARs site location on the Moon's surface. The side-looking direction, despite its modest impact on the imaging swath, plays a dominant role in ascertaining LBSARs imaging region. Furthermore, spatiotemporal variations of ground swath in various epochs were analyzed to emphasize the challenge and importance of characterizing Earth observation capability in the LBSAR.**

***Index Terms*— Imaging region, lunar-based synthetic aperture radar (LBSAR), LBSAR's altitude, spatiotemporal swath, synthetic aperture radar (SAR) configurations.**

NOMENCLATURE

AOL	Argument of latitude.
DE 440	Developed Ephemeris 440.
DEM	Digital elevation model.
DFMR	Doppler frequency modulation rate.
ECEF	Earth-centered Earth-fixed coordinate.
ECI	Earth-centered inertial coordinate.
LBSAR	Lunar-based synthetic aperture radar.
MCI	Moon-centered inertial coordinate.
MCMF	Moon-centered moon fixed coordinate.
MJD	Modified Julian date.
NESZ	Noise equivalent sigma zero.
PRF	Pulse repetition frequency.
RAAN	Right ascension of ascending node.
SAT	Synthetic aperture time.
SNR	Signal-to-noise ratio.
TDB	Barycentric dynamical time.
TOI	Target of interest.

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I. INTRODUCTION

THE Earth observation would benefit significantly from establishing a lunar base owing to its potential to serve as a long-term and high-orbit platform. In this context, the LBSAR can reliably observe Earth with exceptional coverage, which will be attractive for monitoring large-scale geophenomena [1], [2], [3], [4]. Hence, the LBSAR has aroused growing attention since the 2000s [5], [6], [7], [8], [9].

In the synthetic aperture radar (SAR) configurations, the spatial coverage counts for much of Earth observation [10]. The LBSAR might maximize spatial coverage by implementing multiple antennas. However, the implementation comes at the expense of image quality degradation due to the squint effects [11]. Also, potential challenges with constructing and synchronizing multiple antennas on the Moon's surface exist. Therefore, it is preferable to employ a single antenna with zero-Doppler steering at the initial stage of the LBSAR. This practice facilitates acquiring high-quality images and lowering system design difficulties in the SAR systems [12], [13]. Then, the swath that awards the extent of the image scene could be a crucial measure of the LBSARs coverage performance. Furthermore, the rational determination of the swath is a prerequisite for avoiding the range and azimuth ambiguities [14], [15], [16]. Reciprocally, the LBSARs imaging swath is constrained by both imaging ambiguities.

When employing the zero-Doppler steering, the unambiguous swath is further dominated by the zero-Doppler plane [17]. Then, the LBSARs swath is affected by a series of factors, including orbital elements, SAR configurations, and Earth's shape, wherein the orbital elements are susceptible to perturbation forces [18], [19]. In this respect, the orbital perturbations could bring uncertainty to the imaging swath and region, further complicating their spatiotemporal variations [20], [21], [22]. The existing works pertaining to the LBSARs coverage were conducted under the spherical Earth model. Nevertheless, little concern in those studies is paid to the influences of LBSARs configurations, orbital elements, and Earth's ellipsoidal shape on the coverage performance [11], [23].

To tackle the issues above, we probe into the ground swath and quantitatively examine its dependence on the pertinent

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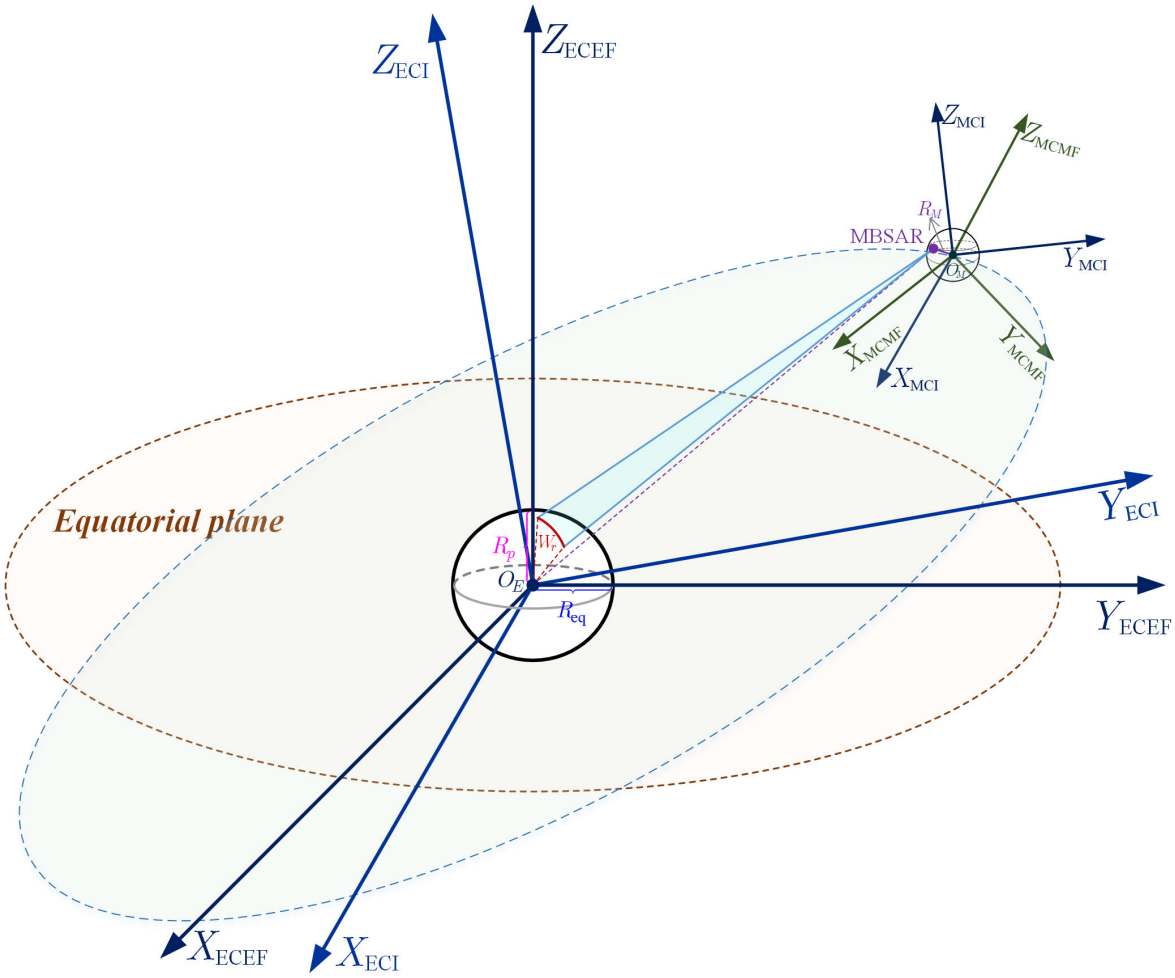


Fig. 1. Earth-observing geometry and ground swath in the LBSAR, where O_E and O_M are centers of Earth and Moon, respectively, R_{eq} and R_p are the equatorial and polar radii of the reference Earth ellipsoid, respectively, R_M is the reference radius of Moon, and W_r is the ground swath of LBSAR.

influencing factors in the LBSAR. We thoroughly explore spatiotemporal variations of imaging swath and region across distinct epochs, demonstrating how orbital perturbation effects affect the coverage performance in the LBSAR. We structure this article as follows. Section II derives the ground and slant swaths and bounds relevant to the constraints of imaging swaths. Section III configures the LBSAR by examining the relevant bounds. Also, this section highlights the quantitative analysis of the ground swath and its sensitivity to the LBSAR configurations. Section IV presents the spatiotemporal variations of the ground swath in various epochs, followed by simulations and assessments of imaging regions. Section V concludes this article.

II. Determination of the Ground Swath and Relevant Bounds in the LBSAR

A. Ground and Slant Swaths in the LBSAR

In the LBSAR, the observation geometric model differs from that of spaceborne SAR: the antenna is placed on the lunar surface where the Earth is observable, at a considerable distance from the barycenter of the platform. Fig. 1 shows the observation geometry of the LBSAR, in which various coordinate systems, including the ECI, ECEF, MCI, and MCMF coordinate systems, are involved. According to Fig. 1, at a specified observation time, the LBSARs position vector in the ECI coordinate is given by

$$\mathbf{R}_{\mathbf{SAR(ECI)}} = \mathbf{R}_{\mathbf{EM(ECI)}} + (R_M + h_{\mathrm{MEL}} + h_{\mathrm{site}})\mathbf{U}_{\mathbf{MCI}}^{\mathbf{MCMF}} \cdot [\cos\Phi_{\mathrm{site}}\cos\Lambda_{\mathrm{site}}, \cos\Phi_{\mathrm{site}}\sin\Lambda_{\mathrm{site}}, \sin\Phi_{\mathrm{site}}]^T \tag{1}$$

where $\mathbf{R}_{\mathbf{EM(ECI)}}$ is the lunar barycenter's position in the ECI, $\mathbf{U}_{\mathbf{MCI}}^{\mathbf{MCMF}}$ is the transformation matrix from MCMF to MCI, R_M is the Moon's reference radius, Φ_{site} is the LBSARs latitude in the MCMF, Λ_{site} is the LBSARs longitude in the MCMF, h_{MEL} is the lunar elevation at the site of $(\Phi_{\mathrm{site}}, \Lambda_{\mathrm{site}})$, and h_{site} is the antenna's altitude of LBSAR.

Although the Moon exhibits an ellipsoidal shape, a slight disparity exists between its polar and equatorial radii [24], [25]. Here, the lunar global DEM shown in Fig. 2, with a reference radius of 1737.4 km, is applied to provide lunar topography, where the ellipsoidal shape of the Moon is implicit in the topography [26].

For a TOI within the LBSARs ground swath, its position in the ECEF coordinate can be written as

$$\mathbf{R}_{\mathbf{TOI(ECEF)}} = \mathbf{U}_{\mathbf{ECEF}}^{\mathbf{ECI}}\mathbf{U}_{\mathbf{ECI}}^{\mathbf{ACS}}\left[\mathbf{R}_{\mathbf{SAR(ACS)}} - \mathbf{R}_{\mathbf{ST(ACS)}}\right] \tag{2}$$

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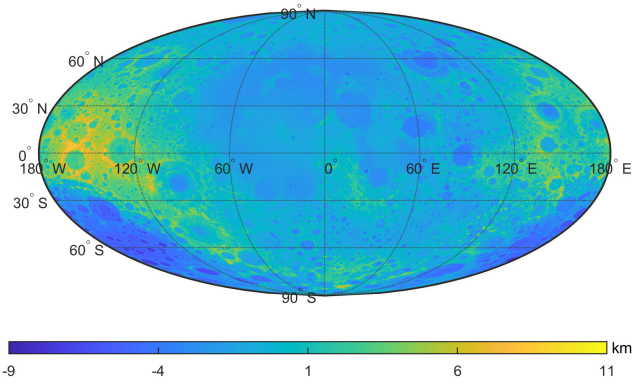


Fig. 2. Global DEM of the Moon with a reference radius of 1737.4 km.

with

$$\mathbf{R}_{\mathbf{SAR(ACS)}} = [-R_{\mathrm{SAR}}, 0, 0]^T \tag{3}$$

$$\mathbf{R}_{\mathbf{ST(ACS)}} = R_{\mathrm{ST}}[-\cos\theta_l, \xi_{\mathrm{sld}} \sin\theta_l \cos\phi_a, \xi_{\mathrm{sld}} \sin\theta_l \sin\phi_a]^T \tag{4}$$

where $\mathbf{U}_{\mathbf{ECEF}}^{\mathbf{ECI}}$ is the transformation matrix from ECI to ECEF, $\mathbf{U}_{\mathbf{ECI}}^{\mathbf{ACS}}$ is the transformation matrix from ACS to ECI, R_{SAR} is the distance from the geocenter to LBSAR, R_{ST} is the slant range of the beam centerline, ϕ_a is the antenna azimuth angle, θ_l is the LBSARs look angle, and ξ_{sld} relates to the side-looking direction, and $\xi_{\mathrm{sld}} = 1$ and $\xi_{\mathrm{sld}} = -1$, respectively, represent the LBSAR looks from the left- and right-hand sides.

In this study, the Moon's orbit and attitude are provided by the DE 440, while the lunar cycle is given by MJD in the time scale of TDB [27]. The associated coordinate transformations can be found in [11], [28], [29], and [30] with additional details. We refrain from repeating a similar analysis.

In (3) and (4), R_{SAR} and R_{ST} take the following forms:

$$R_{\mathrm{SAR}} = \left\| \mathbf{R}_{\mathbf{SAR(ECI)}} \right\|_2 \tag{5}$$

and

$$R_{\mathrm{ST}} = S_1^{-1} \cdot \left[S_2 - \left(S_2^2 - S_1 S_3 \right)^{0.5} \right] \tag{6}$$

where $\| \cdot \|$ represents the ℓ^2-norm product. S_i $(i = 1, 2, 3)$ are functions of ϕ_a and θ_l (antenna azimuth and look angles) and R_{eq} and R_p (equatorial and polar radii of Earth). Their detailed expressions and derivations can be found in the Appendix. When the zero-Doppler steering is employed, the antenna azimuth angle is regulated by the look angle and orbital elements. Then, the slant range R_{ST} is dependent on the look angle of LBSAR in this scenario.

Following (2)–(6), the expressions are given for the position vectors that correspond to the lower and upper boundaries of the beam footprint on Earth as follows:

$$\begin{cases} \mathbf{R}_{\mathbf{LBS}} = \mathbf{U}_{\mathbf{ECEF}}^{\mathbf{ECI}} \mathbf{U}_{\mathbf{ECI}}^{\mathbf{ACS}} \left[\mathbf{R}_{\mathbf{SAR(ACS)}} - \mathbf{R}_{\mathbf{ST(ACS)}}^{\mathbf{near}} \right] \\ \mathbf{R}_{\mathbf{UBS}} = \mathbf{U}_{\mathbf{ECEF}}^{\mathbf{ECI}} \mathbf{U}_{\mathbf{ECI}}^{\mathbf{ACS}} \left[\mathbf{R}_{\mathbf{SAR(ACS)}} - \mathbf{R}_{\mathbf{ST(ACS)}}^{\mathbf{far}} \right] \end{cases} \tag{7}$$

where $\mathbf{R}_{\mathbf{ST(ACS)}}^{\mathbf{near}}$ and $\mathbf{R}_{\mathbf{ST(ACS)}}^{\mathbf{far}}$ are the near and far slant range vectors that correspond to near and far look angles, i.e., θ_l^{near} and θ_l^{far}, respectively. The look angle θ_l relates to θ_l^{near} and θ_l^{far} by

$$\theta_l^{\mathrm{near}} \le \theta_l \le \theta_l^{\mathrm{far}}, \quad \theta_l^{\mathrm{far}} = \theta_l^{\mathrm{near}} + \beta_r \tag{8}$$

with the beamwidth expressed by

$$\beta = k_r \lambda l_r^{-1} \tag{9}$$

where λ is the wavelength of the radar signal, l_r is the aperture length along the range direction, and k_r is the taper factor related to beam steepness [14].

The geocentric angle determining the ground swath is

$$\theta_e = \frac{\mathbf{R}_{\mathbf{LBS}} \mathbf{R}_{\mathbf{UBS}}}{\|\mathbf{R}_{\mathbf{LBS}}\|_2 \|\mathbf{R}_{\mathbf{UBS}}\|_2}. \tag{10}$$

Next, we can determine the slant and ground swaths in the LBSAR. It follows that the slant swath can be obtained by:

$$W_{\mathrm{sr}} = \left\| \mathbf{R}_{\mathbf{ST(ACS)}}^{\mathbf{far}} \right\|_2 - \left\| \mathbf{R}_{\mathbf{ST(ACS)}}^{\mathbf{near}} \right\|_2. \tag{11}$$

Regarding the ground swath, its closed-form solution is unavailable when considering the spatial variability of radius for an ellipsoidal Earth. Thus, a numerical solution is sought; in this event, we can divide the geocentric angle into n subintervals and determine the ground swath as

$$W_r = \frac{\theta_e}{n} \left(R_{\mathrm{TOI}}^{\mathrm{far}} + R_{\mathrm{TOI}}^{\mathrm{near}} + \sum_{k=1}^{n-1} R_{\mathrm{TOI}}(\vartheta_k) \right) \tag{12}$$

with

$$\vartheta_k = k \cdot n^{-1} \theta_e, \quad k = 1, 2, \ldots, n \tag{13}$$

where $R_{\mathrm{TOI}}^{\mathrm{far}}$ and $R_{\mathrm{TOI}}^{\mathrm{near}}$ are the Earth's radius corresponding to the far- and near-look angles, respectively, and $R_{\mathrm{TOI}}(\vartheta_k)$ is the Earth's radius corresponding to ϑ_k. According to (10)–(12), the slant and ground imaging swaths are numerically correlated by the near-look angle and beamwidth.

In the case where the spherical Earth assumption is made, the ground swath can be numerically obtained through (12) by substituting all Earth's radii with the Earth's mean radius or, equivalently, by

$$W_{r(\mathrm{SPH})} = R_E \cdot \theta_{e(\mathrm{SPH})} \tag{14}$$

where R_E is the mean radius of Earth (6371.0 km). The subscript "(SPH)" indicates the spherical Earth assumption.

Given the above framework, we can assess the ground swath and associated imaging region quantitatively as long as the LBSARs site is determined. In what follows, the "site" refers specifically to the LBSARs site on the Moon's surface. The lunar topography is implied in the site location, which is represented by the selenographic coordinate of $(\Phi_{\mathrm{site}}, \Lambda_{\mathrm{site}})$, with "N/S" representing the north/south latitude and "E/W" indicating the east/west longitude, respectively.

B. Constraint of the Swath Versus SAR Configurations

Before proceeding to analyze the LBSAR swath, several critical bounds relevant to the swath and configurations of LBSAR should be clarified. One group of those bounds is decided by the imaging ambiguity. For instance, the radar system with a PRF, f_{PRF}, can only identify Doppler (azimuth) frequencies ranging from $-0.5 f_{\mathrm{PRF}}$ to $0.5 f_{\mathrm{PRF}}$. Otherwise, azimuth aliasing occurs [31]. To refrain from this, one selects that

$$|f_d|_{\max} < 0.5 f_{\mathrm{PRF}} \tag{15}$$

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where f_d is the Doppler frequency, which is spatially varying within the image scene of LBSAR, and it is approximated by

$$f_d \approx f_{\rm dc} + f_{\rm dr}\eta, \quad -0.5T_{\rm SAR} \le \eta \le 0.5T_{\rm SAR} \tag{16}$$

where $f_{\rm dc}$ and $f_{\rm dr}$ are the Doppler centroid and DFMR, respectively. Their expressions can be found in [20] and [32]. $T_{\rm SAR}$ and η are the SAT and azimuth time, respectively.

In the range domain, the received signal must be generated by two pulses emanating from distinct locations on the ground, bringing about differences in their respective slant ranges [33]. This requirement can be expressed as a function of the PRF

$$f_{\rm PRF} < 0.5 \cdot cW_{\rm sr}^{-1} \tag{17}$$

where c is the propagation velocity of the probing signal.

From (15) and (17), the slant swath of the LBSAR is confined to

$$W_{\rm sr} < \frac{c}{4|f_d|_{\max}}. \tag{18}$$

Thus, the swath is limited by imaging ambiguities, especially when the non-Doppler centroid exists in the LBSARs signal. It is preferred to employ the zero-Doppler steering to acquire a wider unambiguous swath. Once the beam center aligns with the zero-Doppler plane, (18) can be given by

$$W_{\rm sr} < 0.5c\left|f_{\rm dr}^{-1}\right|T_{\rm SAR}^{-1}. \tag{19}$$

With (11), (12), and (19), we can analyze the tradeoff between the ground swath and SAT (or, equivalently, the Doppler bandwidth). Also, (9), (11), and (19) give the upper bound of the beamwidth for the unambiguous swath.

Implementing the zero-Doppler steering in LBSAR could introduce constraints. For observing Earth, the beam footprint must illuminate the Earth ellipsoid. The imaging region ought to provide valid echoes with the zero-Doppler centroid. Both requirements restrict the near-look angle within a specific range. Mathematically, the required range can be written as [17]

$$\theta_l^{\rm low} < \theta_l^{\rm near} < \theta_l^{\rm up} \tag{20}$$

with

$$\theta_l^{\rm up} = \sin^{-1}\left(R_p R_{\rm SAR}^{-1}\right) - \beta_r \tag{21}$$

$$\theta_l^{\rm low} = \cot^{-1}\left[\frac{\omega_E R_{\rm SAR}}{|V_{AZ}|}\sqrt{\left(\frac{V_{AX}}{\omega_E R_{\rm SAR}} - \cos i\right)^2 + \sin^2 i\cos^2 u}\right] \tag{22}$$

where ω_E is the Earth's rotational angular velocity, i is the orbital inclination in the ECI, u is the AOL in the ECI, V_{AX} is the x-component of $\mathbf{V}_{\mathbf{SAR(ACS)}}$, and V_{AZ} is the z-component of $\mathbf{V}_{\mathbf{SAR(ACS)}}$.

It is noteworthy that the upper bound of the near-look angle is determined by the polar radius, the minimum radius of Earth. The LBSARs velocity vector in the ACS, $\mathbf{V}_{\mathbf{SAR(ACS)}}$, is given in [32]. The detailed derivation regarding the bounds of near-look angle can be found in [17].

By far, we have presented the swath and its pertinent bounds versus LBSARs configurations. In Section III, we shall quantitatively probe into the swath and its sensitivity to the configurations of the LBSAR.

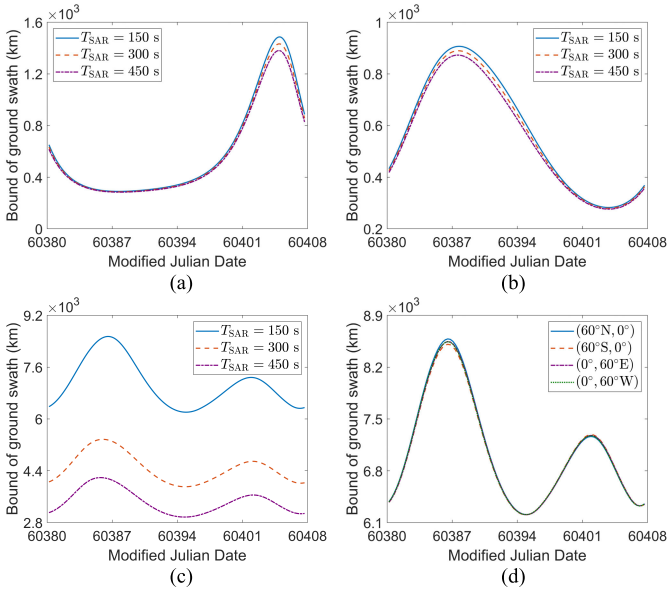


Fig. 3. Within one lunar cycle. The bound of ground swath at the site of (0°, 0°) with various SATs under antenna azimuth angles of (a) 45° and (b) 135°. The bound of ground swath in the zero-Doppler steering mode at the site location of (0°, 0°) (c) with various SATs and (d) with an SAT of 150 s at different site locations.

III. Configuration and Sensitivity Analysis of the Ground Swath in the LBSAR

This section investigates the ground swath and its sensitivity versus the LBSARs configurations. To proceed with this study, we choose one lunar cycle that spans from MJD 60 380.21214196 to MJD 60 407.67627033. The carrier frequency and the antenna's altitude in LBSAR are set to 1.2 GHz and 1.5 m, respectively. To properly configure the LBSAR, we inspect the bounds relevant to the imaging swath.

A. Selection of the LBSAR Configurations

Note that the relevant bounds exhibit similar patterns in both side-looking directions; for simplicity, the LBSAR is set to left-looking. Given the prerequisites above, we first inspect the bounds of the ground swath aroused by imaging ambiguity. Fig. 3 shows the upper bound for the ground swath in the squint-looking and zero-Doppler steering modes under various SATs. When the zero Doppler steering is employed, the bounds of the ground swath with an SAT of 150 s at different site locations are shown in Fig. 3. In all those cases, the near-look angle is set to 0.2°.

It is observed from Fig. 3 that in the squint-looking mode, the upper bound of the ground swath is mainly determined by the antenna azimuth angle, whereas the SAT exerts a relatively small impact on the ground swath. Thus, in the squint-looking mode, the unambiguous ground swath is limited to a value below 1500 km or even smaller. When employing the zero-Doppler steering, the DFMR and SAT dominate the upper bound of the ground swath, while the LBSARs site location shows a slight impact on this bound. Moreover, a promising wider unambiguous ground swath is available, especially when a shorter SAT, e.g., 150 s, is applied in the LBSAR. In this case, the SAT provides an azimuth resolution on the decameter

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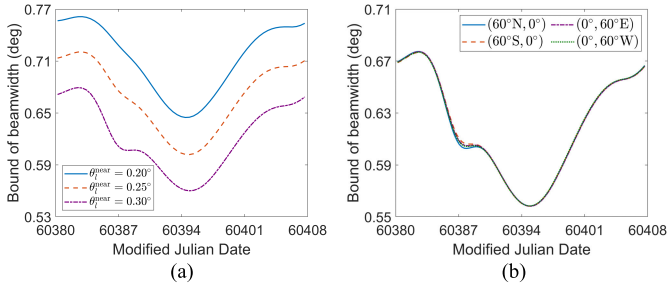


Fig. 4. Within one lunar cycle. (a) Bounds of beamwidth with various near-look angles at the site location of (0°, 0°). (b) Bounds of beamwidth at various sites with near-look angles of 0.30°.

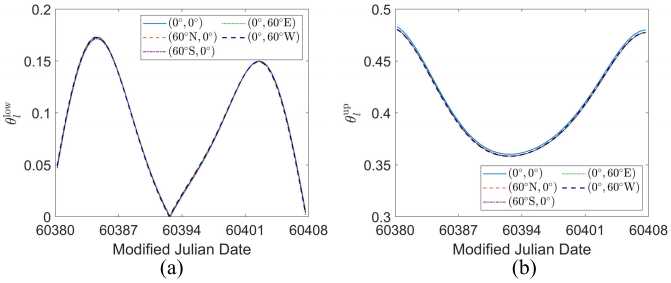


Fig. 5. Within one lunar cycle. The bounds of near-look angles at various sites: (a) lower bound and (b) upper bound.

level [32], [34], helping to mitigate the defocusing phenomena caused by the atmosphere and orbital perturbations [20], [21], [22], [35], [36]. On the other hand, the azimuth resolution on the metric level or finer seems less needed in the LBSAR as it primarily focuses on observing the large-scale geophenomena [37]. Consequently, an SAT of 150 s is selected for the LBSAR to avoid image ambiguity.

Next, using the SAT of 150 s, we determine the beamwidth for the unambiguous swath in the LBSAR. With this end in view, the bounds for the beamwidth at the site of (0°, 0°) and those with a fixed near-look angle (i.e., 0.3°) at different site locations are plotted in Fig. 4. We observe that the threshold of beamwidth varies along the LBSARs orbit. Its magnitude is also impaired by the site location of LBSAR, but to a minor extent. In addition, the beamwidth's threshold increases with the increasing near-look angle, implying that the near-look angle and beamwidth bear a coupling restraint relationship. Noticeably, the threshold serves as an upper bound for the beamwidth, necessitating that the beamwidth remains constrained below the minimum threshold (0.558°) within one lunar cycle. Hence, the beamwidth of LBSAR is set to 0.54°. The beamwidth is correlated to the taper factor and real aperture length along the range direction. For a taper factor of 1.0, the chosen beamwidth corresponds to a real aperture length of 26.53 m for the LBSAR.

Next, we examine the low and upper bounds of the near-look angle using a beamwidth of 0.54°. Fig. 5(a) and (b) presents both bounds at various sites. It is observed that the lower and upper bounds of near-look angle vary temporally. In addition, the LBSARs site location could affect the lower bound to a diminutive degree, while it exerts a relatively more noticeable impact on the upper bound. Furthermore, we find that the

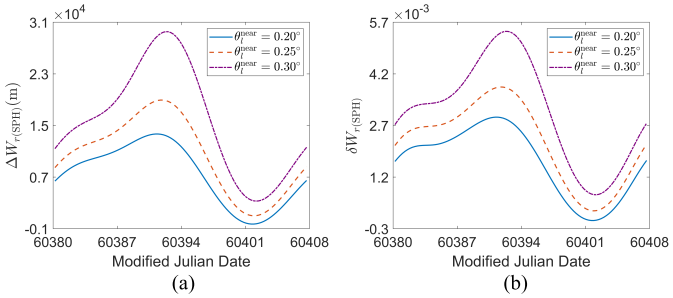


Fig. 6. Within one lunar cycle. The error associated with the ellipsoidal shape of Earth under various near-look angles: (a) $\Delta W_{r(\mathrm{SPH})}$ and (b) $\delta W_{r(\mathrm{SPH})}$.

lower bound of the near-look angle is consistently smaller than 0.20°, whereas the minimal upper bound exceeds 0.36° within one lunar cycle regardless of LBSARs site location. However, Fig. 4 suggests a near-look angle larger than 0.30° might bring about imaging ambiguity for the LBSAR with the given beamwidth. Therefore, we select the range for the near-look angle to be between 0.20° and 0.30°.

B. Ground Swath Analysis

Section II presents that the LBSARs imaging swath is coupling influenced by several factors. In this section, the quantitative assessment of the ground swath and its dependence on the influencing factors are clarified.

1) Swath's Dependence on Earth Ellipsoid: To account for the spatial variability of Earth's radius on the ground swath with respect to the spherical Earth model, we define two associated errors

$$\Delta W_{r(\mathrm{SPH})} = W_r - W_{r(\mathrm{SPH})}, \quad \delta W_{r(\mathrm{SPH})} = W_r^{-1} \cdot \Delta W_{r(\mathrm{SPH})}. \tag{23}$$

Here, we adopt the WGS 84 reference ellipsoid, with a semimajor axis of 6378.137 km and an inverse flattening of 298.257223563 [38].

Referring to the LBSARs site at the selenographic coordinate of (0°, 0°), we present the errors $\Delta W_{r(\mathrm{SPH})}$ and $\delta W_{r(\mathrm{SPH})}$ under various near-look angles in Fig. 6. It is evident that both errors show temporal variations within one lunar cycle. Moreover, the magnitude of each error tends to escalate as the near-look angle increases. In the case where a near-look angle of 0.3° is used, the error $\Delta W_{r(\mathrm{SPH})}$ varies between several and tens of kilometers, depending on the orbital position of LBSAR. Correspondingly, the magnitude of relative error $\delta W_{r(\mathrm{SPH})}$ spans from 6.87×10^{-4} to 5.44×10^{-3}. From this perspective, the adoption of the spherical Earth model remains valid only when a tolerable relative error of 5.44×10^{-3} or, correspondingly, a tolerable variation of 30 km in the ground swath, is acceptable. Otherwise, the spatial variability of Earth's radius given rise by the ellipsoidal shape must be considered.

2) Swath's Dependence on Side-Looking Direction: Due to the ellipsoidal shape of Earth and the perturbed orbit of LBSAR, the Earth's radius and geocentric angle within the image scene exhibit disparities between the left- and right-looking LBSARs. As a result, the ground swath of the LBSAR

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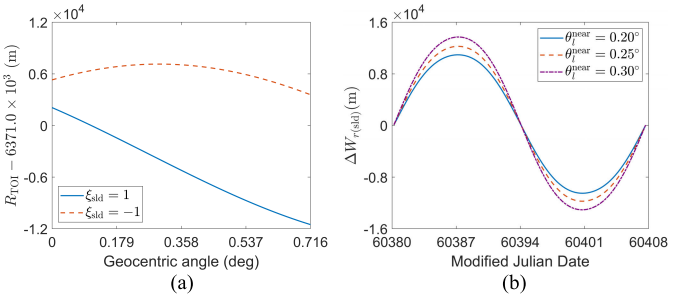


Fig. 7. In the LBSAR, (a) Earth's radius versus geocentric angle under a near-look angle of 0.2° at MJD 60 384.0. (b) Within one lunar cycle, the error $\Delta W_{r(\mathrm{sld})}$ under various near-look angles.

is potentially influenced. To examine the impact of side-looking direction on the ground swath, two errors are proposed as

$$\Delta W_{r(\mathrm{sld})} = W_{r(\xi_{\mathrm{sld}}=1)} - W_{r(\xi_{\mathrm{sld}}=-1)} \tag{24}$$

where $W_{r(\xi_{\mathrm{sld}}=1)}$ and $W_{r(\xi_{\mathrm{sld}}=-1)}$ are the ground swaths of left- and right-looking LBSARs, respectively.

Fig. 7 compares the relationships regarding the Earth's radius versus the geocentric angle between the left- and right-looking LBSARs at MJD 60 384.0. Then, the errors $\Delta W_{r(\mathrm{sld})}$ under various near-look angles are also shown in Fig. 7 to demonstrate the influence of side-looking direction on the ground swath.

As shown in Fig. 7, the left- and right-looking LBSARs give rise to two distinct relationships between the Earth's radius and geocentric angle within the image scene, depending on the location of LBSAR along its orbit. As a result, the ground swath exhibits discrepancies under different side-looking directions unless the LBSAR is located at the ascending or descending nodes. The errors associated with these discrepancies, namely, $\Delta W_{r(\mathrm{sld})}$, show upward tendencies with the increasing near-look angle. In particular, when a near-look angle of 0.3° is employed, the error $\Delta W_{r(\mathrm{sld})}$ fluctuates between a range of several to tens of kilometers. Consequently, the influence of side-looking direction should be taken into account, especially when the near-look angle is larger.

3) Swath's Dependence on Near-Look Angle: Fig. 8 demonstrates the ground swath versus the near-look angle within one lunar cycle, where the LBSAR is assumed to be located at the site of (0°, 0°). It is observed that the ground swath is subject to temporal variation. This temporal variability is sensitive to the LBSARs near-look angle: an apparent upward trend appears in the ground swath as the near-look angle increases, irrespective of the orbital position of LBSAR. In this event, the ground swath consistently surpasses 3800 km for a near-look angle of 0.2°. In contrast, the minimal ground swath can exceed 4000 km in the LBSAR when a near-look angle of 0.3° is employed.

4) Swath's Dependence on Site Location: The ground swath might be affected by the site location of LBSAR on the Moon's surface. The errors defined in (25) can be used to assess the influence of this factor

$$\Delta W_{r(\mathrm{site})}(\Phi_{\mathrm{site}}, \Lambda_{\mathrm{site}}) = W_r(\Phi_{\mathrm{site}}, \Lambda_{\mathrm{site}}) - W_r(0°, 0°) \tag{25}$$

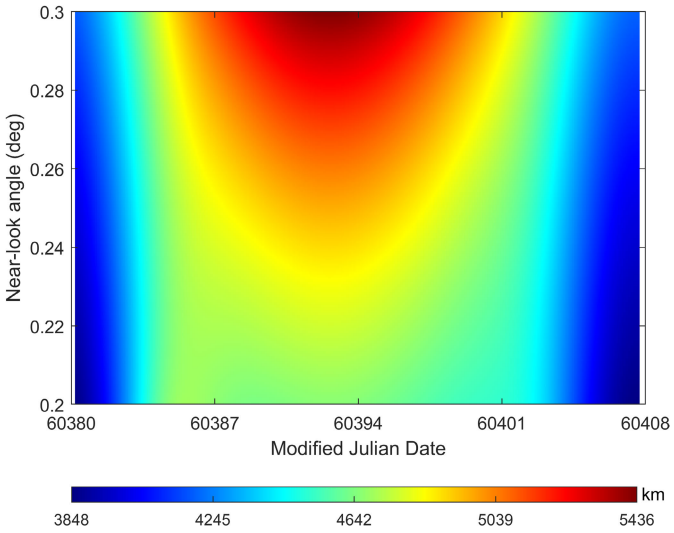


Fig. 8. Ground swath versus the near-look angle within one lunar cycle.

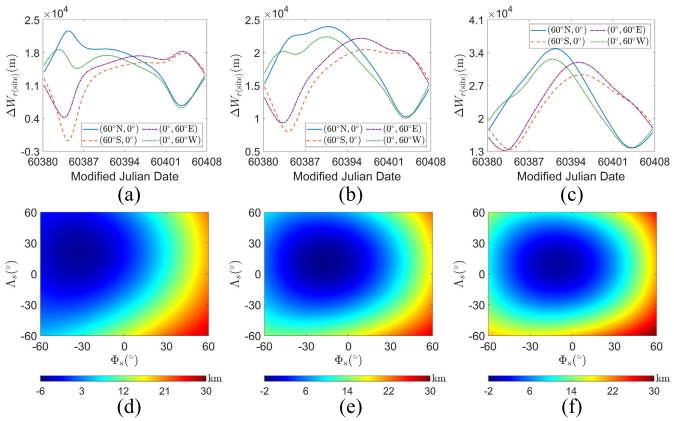


Fig. 9. Error associated with the impact of site location under various conditions, where (a) and (d) first, (b) and (e) second, and (c) and (f) third columns represent the near-look angles of 0.20°, 0.25°, and 0.30°, respectively, while (a)–(c) first and (d)–(f) second rows indicate the errors $\Delta W_{r(\mathrm{site})}$ at various site locations within one cycle and that versus site locations at MJD 60 384.0, respectively.

where $W_r(\Phi_s, \Lambda_s)$ and $W_r(0°, 0°)$ are the ground swaths of the LBSAR at the sites of $(\Phi_{\mathrm{site}}, \Lambda_{\mathrm{site}})$ and (0°, 0°), respectively.

Fig. 9 presents the error $\Delta W_{r(\mathrm{site})}$ at various site locations under various near-look angles within one lunar cycle. Also, Fig. 9 shows the error $\Delta W_{r(\mathrm{site})}$ versus site locations at the instant of MJD 60 384.0. As seen, the magnitudes and distributions of the error $\Delta W_{r(\mathrm{site})}$ are correlated to the site location on the Moon's surface, which also exhibits temporal variations. Furthermore, a larger near-look angle exalts much greater impacts on the sensitivity of the ground swath to site location. The above phenomena indicate that the impact of site location interconnects with all other factors when considering the dependence of the ground swath on it. The variation in ground swath due to the influence of site location can be up to several dozen kilometers. In this regard, care should be taken when selecting the site location of the LBSAR, especially if a wider ground swath is required.

This section discusses the ground swath and its dependence on the variability of Earth's radius's variability and

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TABLE I
CORRESPONDING EPOCHS FOR THE CYCLES OF THE LBSAR IN MJD

Cycles	T_{lc1}	T_{lc2}	T_{lc3}
Epoch in MJDs	60380.21214196 ~60407.67627033	60407.67627033 ~60435.06387225	60435.06387225 ~60462.33321370
Cycles	T_{lc4}	T_{lc5}	T_{lc6}
Epoch in MJDs	63767.76549253 ~63795.04618845	63795.04618845 ~63822.38548058	63822.38548058 ~63849.80026437

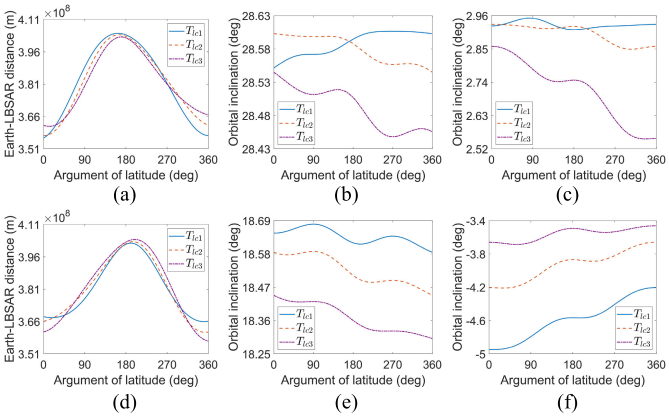


Fig. 10. Variations of orbital elements across various cycles, where (a) and (d) first, (b) and (e) second, and (c) and (f) third columns represent Earth--Moon distance, orbital inclination, and RAAN, respectively, while (a)–(c) first and (d)–(f) second rows stand for cycles of T_{lc1}–T_{lc3} and T_{lc4}–T_{lc6}, respectively.

LBSARs near-look angle, side-looking direction, and site location, whose influences are coupled. Another critical issue concerning the ground swath is the orbital variation, which is subject to the orbital perturbation effects. Consequently, the spatiotemporal variation of the ground swath and relevant imaging region are persistently affected, to be detailed in Section IV.

IV. SPATIOTEMPORAL VARIATION OF THE GROUND SWATH IN DIFFERENT CYCLES OF THE LBSAR

A. Spatiotemporal Ground Swath in the LBSAR

In the LBSAR, the perturbation effects induce continuous variation in the orbital elements of LBSAR. Due to such effects, the orbital inclination is subject to periodical variation. Thus, two extreme cases under which the orbital inclination approaches maximal and minimal values are considered in this study. Three consecutive cycles under each case are selected, and the epochs in MJD corresponding to selected cycles are listed in Table I. Fig. 10 presents the orbital elements of LBSAR as a function of AOL in those selected cycles. In this case, the LBSAR is located at the selenographic coordinate of (0°, 0°). As the semimajor axis, eccentricity, and AOP are implicitly contained in Earth-LBSAR distance, we present the variations of Earth-LBSAR distance across those cycles for simplicity.

Fig. 10 elucidates that each orbital element exhibits distinct patterns over time. When the zero-Doppler steering is applied,

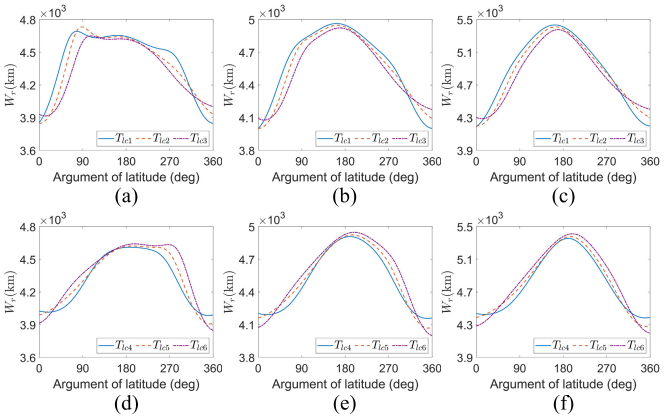


Fig. 11. Ground swath of the LBSAR as a function of AOL across various cycles, where (a) and (d) first, (b) and (e) second, and (c) and (f) third columns represent near-look angles of 0.20°, 0.25°, and 0.30°, respectively, while (a)–(c) first and (d)–(f) second rows stand for ground swaths in cycles of T_{lc1}–T_{lc3} and T_{lc4}–T_{lc6}, respectively.

all temporal-varying orbital elements can have a coupled effect on the zero-Doppler beam pointing, particularly during distinct cycles. Therefore, there would be uncertainties in the spatiotemporal ground swath. Moreover, it is arduous, if not impossible, to provide a general periodic characteristic for these elements, adding to, in effect, the complexity of the spatiotemporal variation in the LBSARs imaging swath. To provide a criterion for measuring diversities of spatiotemporal ground swath between two distinct cycles, we may define a relative error as

$$\delta W_{r(\mathrm{cy})}(T_{lci}, T_{lcj}; u) = \frac{W_r(T_{lci}, u) - W_r(T_{lcj}, u)}{W_r(T_{lci}, u)} \tag{26}$$

where $W_r(T_{lci}, u)$ and $W_r(T_{lcj}, u)$ are the ground swaths at the AOL of u in cycles T_{lci} and T_{lcj}, respectively.

Without loss of generality, the LBSAR that looks from the left-hand side is located at the site of (0°, 0°). The ground swaths with various near-look angles in epochs T_{lc1}–T_{lc3} and T_{lc4}–T_{lc6} against the AOL are presented in Fig. 11. To evaluate the fluctuation in ground swaths across different cycles, we depict the associated relative errors of ground swaths between these distinct cycles in Fig. 12.

Figs. 11(a)–(c) and 12(a1)–(a3) show that within an epoch of T_{lc1}–T_{lc3}, different cycles in the same AOL correspond to distinct ground swaths. The rationale behind this phenomenon is attributed to the orbital perturbations. In addition, the near-look angle could exert an impact on the ground swath and its spatiotemporal variations: For a near-look angle of 0.3°, the maximal variation in the ground swath can reach over 1000 km within one cycle. Furthermore, the relative error $\delta W_{r(\mathrm{cy})}$ shows diverse variability at different AOLs, the degree of which is also dependent on the near-look angle. As a result, it is possible to observe a maximal fluctuation of 5.37% in the ground swath between two consecutive cycles. Furthermore, the longer the interval between two inconsecutive cycles, the stronger the fluctuation of the ground swath, suggesting that the accumulation of orbital perturbation exerts a more powerful impact on the spatiotemporal ground swath.

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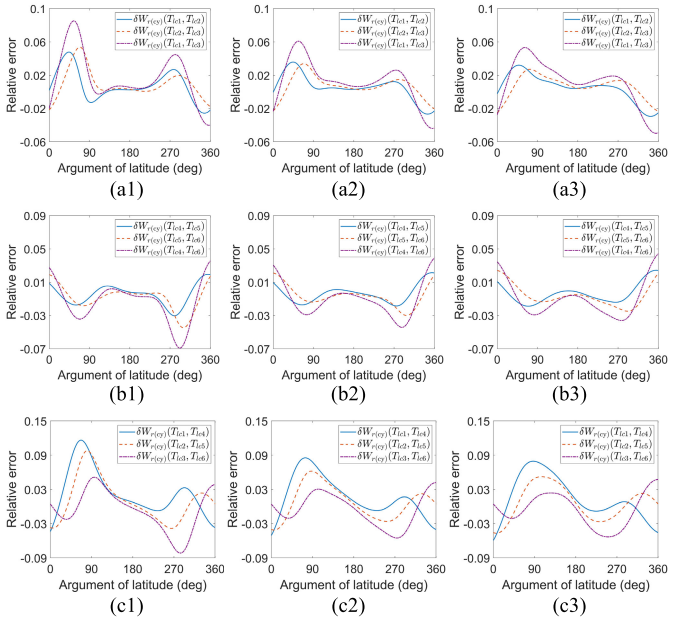


Fig. 12. Relative error of ground swath between two distinct cycles, where (a1)–(c1) first, (a2)–(c2) second, and (a3)–(c3) third represent near-look angles of 0.20°, 0.25°, and 0.30°, respectively, while (a1)–(a3) first and (b1)–(b3) second rows indicate the relative errors in cycles of T_{lc1}–T_{lc3} and T_{lc4}–T_{lc6}, respectively, and (c1)–(c3) third row stands for relative error between cycles T_{lc1}–T_{lc3} and T_{lc4}–T_{lc6}.

Figs. 11(d)–(f) and 12(b1)–(b3) illustrate that in epochs T_{lc4}–T_{lc6}, the ground swath exhibits divergent variation tendencies compared to those observed in epochs T_{lc1}–T_{lc3}. The relevant fluctuations also present different regular patterns in this epoch. There are, however, some similarities between the ground swaths in both epochs: the spatiotemporal ground swath and corresponding fluctuations are on the same level under the same configurations. From there, a wider ground swath is made possible by the LBSAR with a larger near-look angle. Between two gaped cycles, the fluctuation in the ground swath, which is strongly associated with the near-look angle, is more pronounced than that between two consecutive cycles. In this regard, the orbital perturbation induces aperiodic fluctuation in the ground swath across different epochs. More significantly, the accumulation of orbital perturbations more markedly influences the spatiotemporal ground swath and its dependence on configurations in the LBSAR.

The secular Earth observation is one of the foremost missions in the LBSAR; the fluctuation in the ground swath between two cycles with a long-time interval is more deserving of attention. In Fig. 12(c1)–(c3), we show that after approximately 9.3 years, the fluctuation in the ground swath, which is still sensitive to the near-look angle, becomes stronger at most locations with irregular variation patterns. For instance, between the cycles T_{lc1} and T_{lc4}, the maximal fluctuation appears at the AOL of 67.05°, with a magnitude of 11.64% for a near-look angle of 0.20°. By contrast, when implementing a near-look angle of 0.30°, the peak of fluctuation shifts to the AOL of 86.87°, while its magnitude diminishes to 7.94%. In both cases, the fluctuations are in consequence of substantial variation in the ground swath spanning hundreds of kilometers. The above phenomena arise due to cumulative influences exerted by the LBSARs orbital elements, each undergoing varying degrees of temporal variations under perturbation effects. Therefore, the orbital perturbation could inevitably affect the coverage performance, which deserves special care in the LBSAR Earth observation.

We have probed into the spatiotemporal ground swath and associated fluctuations across different epochs. Another critical issue concerning the coverage performance rests with imaging regions of the LBSAR, to be elaborated on next.

B. Imaging Region by the LBSAR

The ground swath of the LBSAR could be affected by its site location on the Moon's surface, and it is critical to inspect how the site location impacts the imaging region in the LBSAR. Referring to an observation time of 150 s that begins from MJD 60 380.21214196, Fig. 13 compares the imaging region of the LBSAR at the selenographic coordinate of (0°, 0°) to that at other selenographic coordinates.

Fig. 13 shows the imaging region at the selenographic coordinate of (0°, 0°) differs, to some extent, from that at other sites, though there are prominent overlap regions. In particular, respecting the selenographic coordinate of (0°, 0°), the site at the Moon's northern latitude tends to shift the imaging region toward the northwest and vice versa. Suppose that the LBSAR is located in the Moon's western hemisphere, the imaging region tends to drift to the northeast. The shifting is least pronounced for a site in Moon's eastern longitude. Furthermore, the comparison of Fig. 13(a)–(d) to (e)–(h) shows that the site location-induced coverage shifting deteriorates at a larger near-look angle. Therefore, it is clear that the selection of site location has a profound effect on the ground swath and associated imaging region. Assured attention must be paid to the site selection when deploying the LBSAR.

Next, referring to the site at the selenographic coordinate of (0°, 0°), Fig. 14 simulates the coverage of the left- and right-looking LBSARs under various near-look angles within one day corresponding to the epoch from MJD 60 380.21214196 to MJD 60 381.21214196.

It is shown that the imaging region varies along with the LBSAR motion relative to Earth (from west to east), the extent of which is regulated by the LBSAR configurations. Specifically, the lower boundary of the imaging region is related to the near-look angle, while the upper boundary is co-determined by the beamwidth and near-look angle. Therefore, the larger the near-look angle, the more extensive coverage can be. A larger near-look angle also makes it possible to imagine the relatively high latitudinal regions on Earth. Regarding the impact of side-looking direction, there is a stark contrast in the corresponding imaging regions, with left-looking LBSAR mainly imaging the northern hemisphere and the right-looking one focusing on the southern hemisphere of Earth.

We examine the imaging regions of left-looking LBSAR in two individual cycles with the site located at (0°, 0°) in the selenographic coordinate. As extensive coverage is more desirable in the LBSAR, the near-look angle is configured

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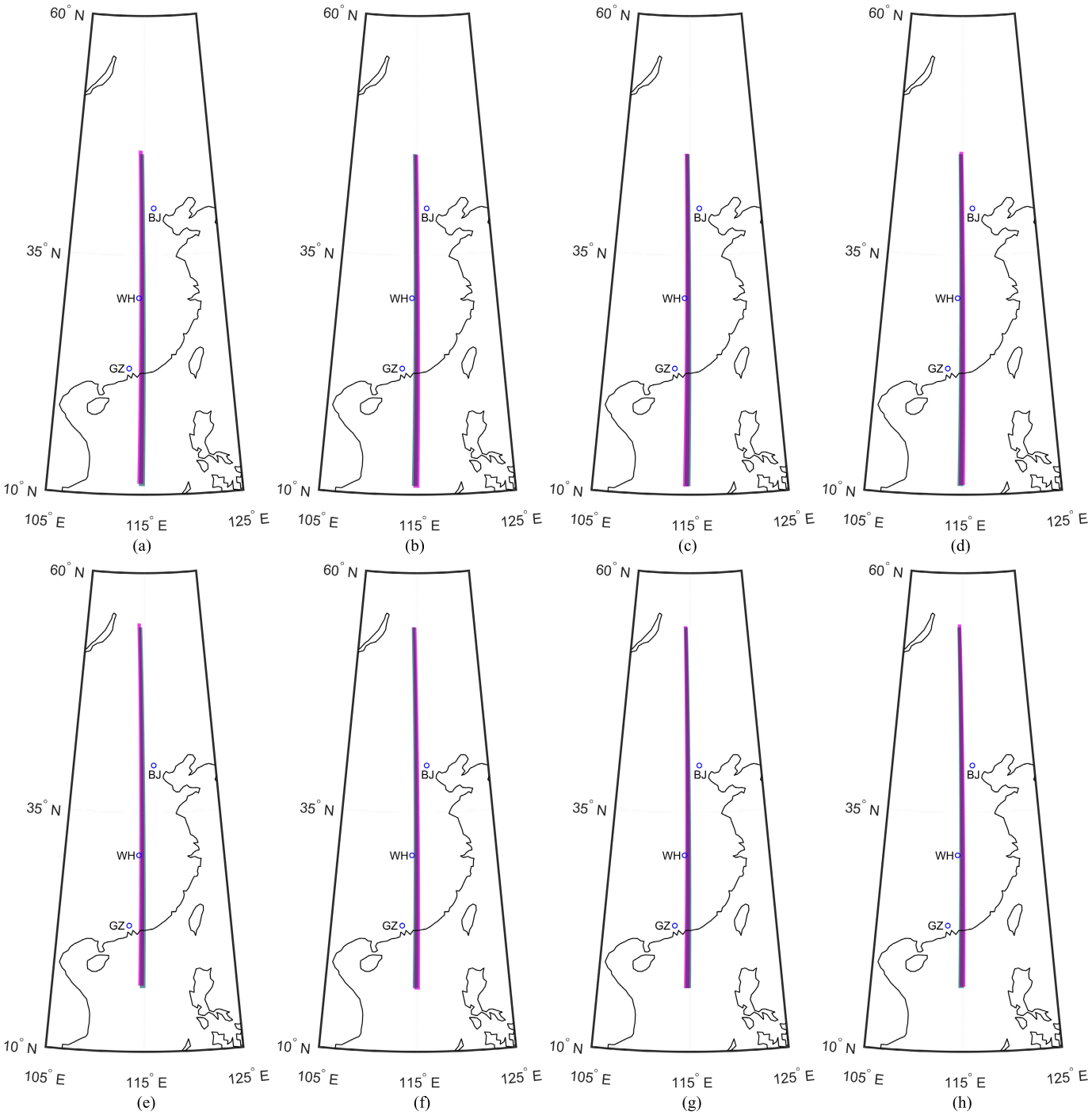


Fig. 13. Under a near-look angle of 0.2°, the imaging region of LBSAR at the selenographic coordinate of (0°, 0°) versus that at selenographic coordinates of (a) (60°N, 0°), (b) (60°S, 0°), (c) (0°, 60°E), and (d) (0°, 60°W); under a near-look angle of 0.3°, the imaging region of LBSAR at the selenographic coordinate of (0°, 0°) versus that at selenographic coordinates of (e) (60°N, 0°), (f) (60°S, 0°), (g) (0°, 60°E), and (h) (0°, 60°W), where the dark cyan region represents the imaging region of LBSAR at the selenographic coordinate of (0°, 0°), the magenta region means the imaging region of LBSAR at another site on the Moon's surface, and the dark purple region stands for the overlap region. "GZ," "WH," and "BJ" stand for Guangzhou, Wuhan, and Beijing, respectively.

as 0.30°. Fig. 15 presents the coverage of LBSAR with the cycle T_{lc1} when the orbital inclination is around its peak. To evaluate the LBSARs repeat observation performance, we select three test locations on 113°E longitude, designated by TOI_1, TOI_2, and TOI_3 with latitudes being 15°N, 45°N, and 70°N, respectively.

Fig. 15 states clearly that the left-looking LBSAR could cover most regions of the northern hemisphere in the cycle T_{lc1}, while only a limited area in the southern hemisphere can be observed. Besides, the LBSARs repeat observation is irregular, depending upon the Earth's latitude. For example, TOI_1 at the low latitude could be observed from March 27,

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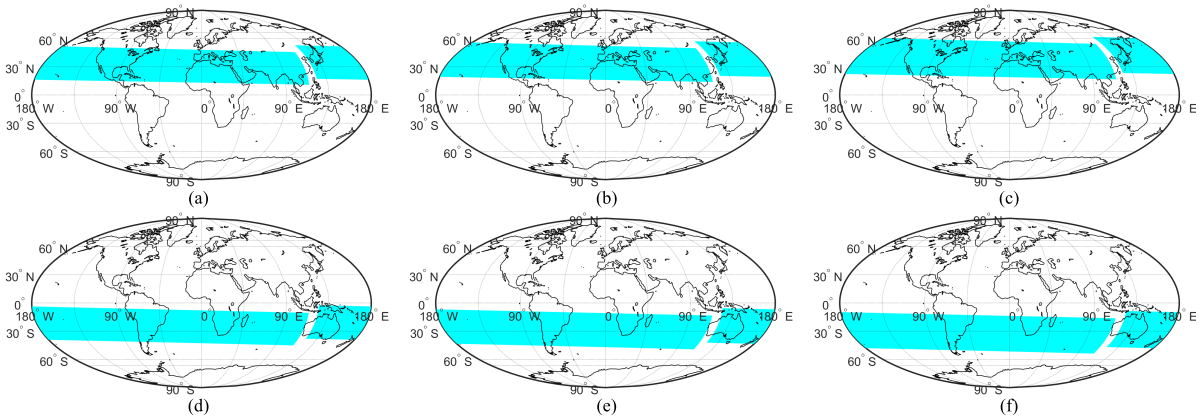


Fig. 14. Imaging region of the LBSAR within a day (24 h), where (a) and (d) first, (b) and (e) second, and (c) and (f) third columns stand for daily coverages with near-look angles of 0.20°, 0.25° and 0.30°, respectively, while (a)–(c) first and (d)–(f) second rows represent coverages of left- and right-looking LBSARs, respectively.

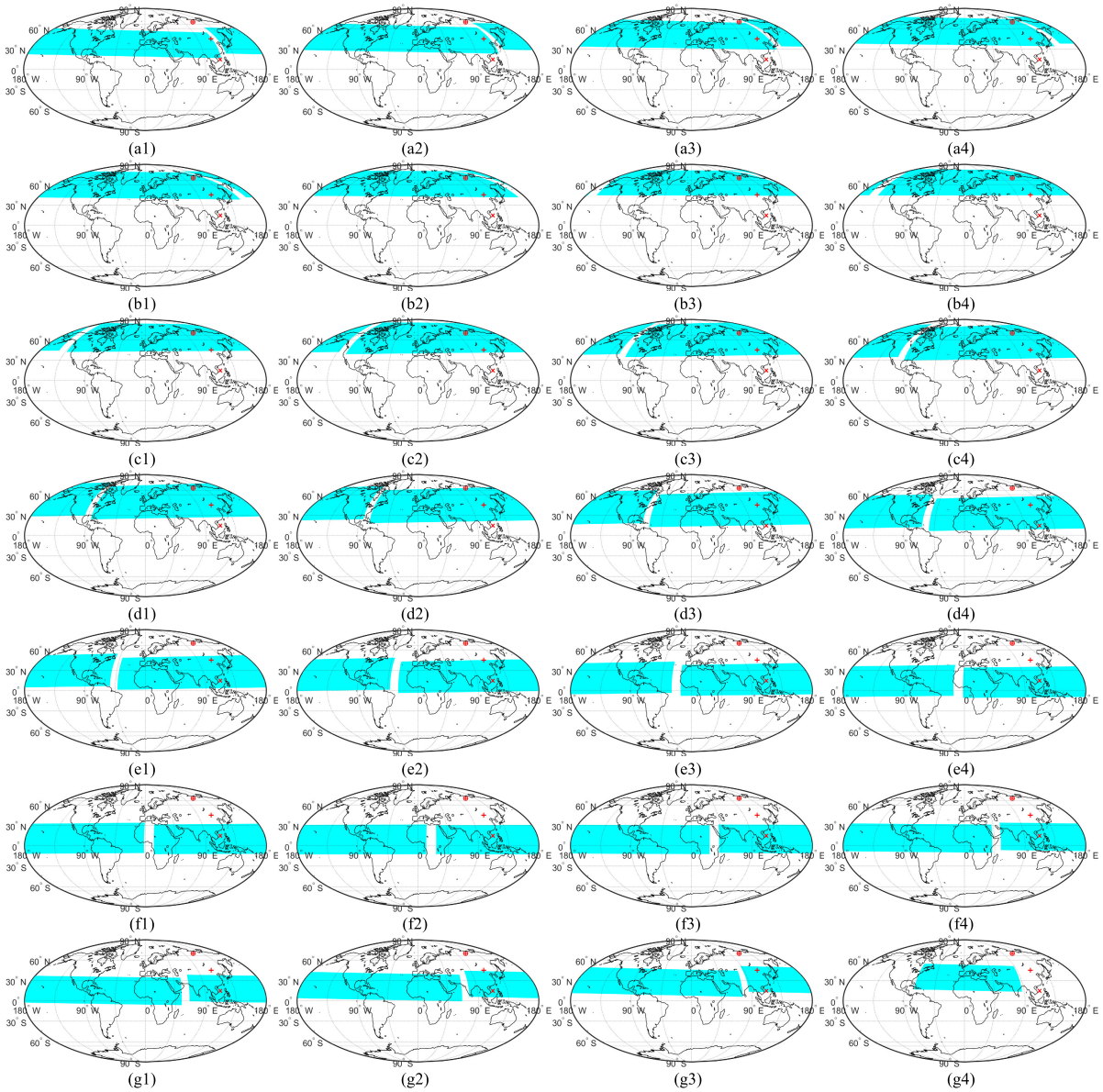


Fig. 15. Imaging region of left-looking LBSAR with a near-look angle of 0.3° within the cycle T_{lc1}, where the symbols "×," "+," and "⊕" stand for the TOIs located at low, middle, and high latitudes, respectively. Each graph (a1)–(g3) represents a daily coverage except for the last one (g4), which stands for the coverage from 05:05:29, April 7, 2024, to 16:13:50, April 7, 2024.

2024 to April 6, 2024, but it is not feasible to be detected by the LBSAR during other parts of cycle T_{lc1}. By contrast, it is feasible to observe TOI_3 approximately once a day from March 14, 2024 to March 23, 2024. In the high latitude,

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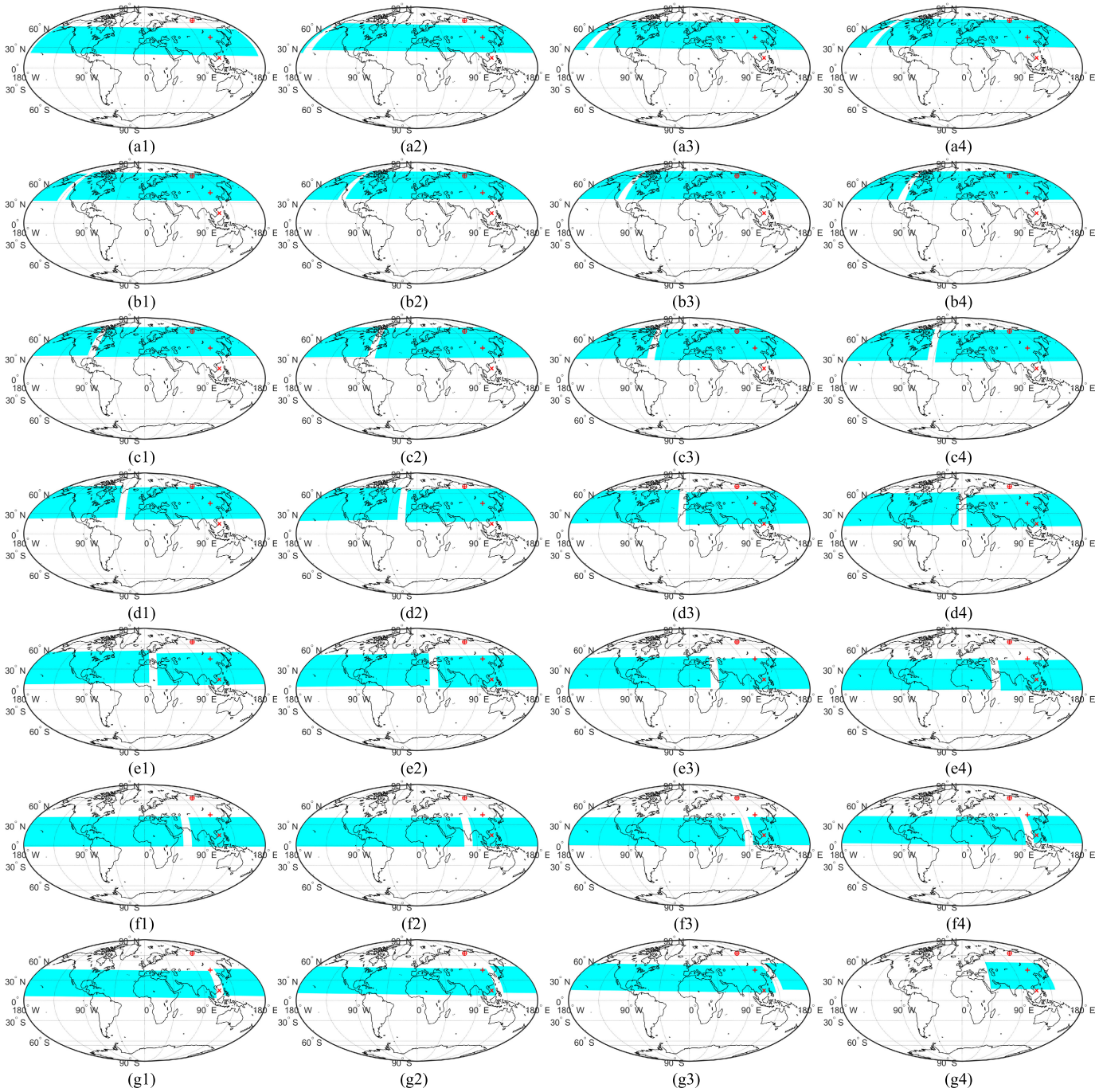


Fig. 16. Imaging region of left-looking LBSAR with a near-look angle of 0.3° within the cycle T_{lc4}, where the symbols "×," "+," and "⊕" stand for the TOIs located at low, middle, and high latitudes, respectively. Each graph (a1)–(g3) represents a daily coverage except for the last one (g4), which stands for the coverage from 18:22:19, July 16, 2033, to 01:06:30.68, July 17, 2033.

especially the polar region, the higher the Earth's latitude, the less frequently it is observed by the LBSAR. Interestingly, TOI_2 can be repeatedly observed 16 times in the selected cycle, indicating that at the mid-latitude, the major inhabited region on Earth can be observed mostly.

We inspect the spatiotemporal imaging region under the small orbital inclination by selecting the cycle T_{lc4}. Fig. 16 presents the daily coverage of left-looking LBSAR in the selected cycle. We see that the LBSAR cannot observe parts of the Arctic region in the selected cycle. Concurrently, the region in the southern hemisphere observed by the LBSAR declines sharply. Consequently, the LBSAR stretches across a narrower range of latitudes within one cycle under the small orbital inclination. The repeat observation also presents divergent trends compared to that under the large orbital inclination. For instance, the quantitative evaluation shows that TOI_1 and TOI_3 are, respectively, observed 11 and 9 times, while the observation of TOI_2 is repeated 22 times in the cycle T_{lc4}. Accordingly, when the orbital inclination is small, the repeat observation of low latitude diminishes

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within one cycle, as does high latitude. In contrast, more repeat observations of mid-latitude are available by the LBSAR.

From the results above, we conclude that the LBSAR can successively observe the Earth, making seamless coverage on a hemispheric scale feasible within one cycle. However, the repeat observation of a specified Earth region, where the geographical position and orbital perturbations exert, might be disrupted and even irregular in different cycles. Hence, the repeat period, either right or left looking, is difficult, if not impossible, to reckon in an Earth-observing LBSAR system.

V. Conclusion

This article analyzes the dependences of imaging swaths on the Earth's ellipsoidal shape, radar configurations, and site location in the LBSAR. Considering the temporal-varying perturbed orbital elements, it also quantitatively evaluates the spatiotemporal imaging swath and associated regions. We find that the ground swath is dependent on the beamwidth, near-look angle, and orbit of the LBSAR, and to a lesser extent on the Earth's ellipsoidal shape, LBSARs site location, and side-looking direction. Hence, the more extensive ground swath can be achieved with a larger near-look angle. Moreover, the larger near-look angle makes it more feasible to observe the high-latitude regions on Earth. Notably, the imaging region is also susceptible to side-looking direction, making the left-looking LBSAR well-suited for monitoring the northern hemisphere and vice versa. The LBSARs orbit elements, influenced by the perturbation effects, could bring about aperiodic variations in the spatiotemporal ground swath and associated imaging region. As a result, discontinuous and irregular imaging swath in the long-term observation of a specified region inherently prevails. Nevertheless, the LBSAR offers a ubiquitous swath coverage over thousands of kilometers, which is favorable for observing Earth.

Before closing, it is noteworthy that the SNR and NESZ are highly correlated to the imaging swath. These parameters may be influenced by the spatiotemporal variation of ground swath. We will pursue this exciting yet important topic in our future study.

Appendix
Derivation of Slant Range Under the Reference Earth Ellipsoid

From (2), the position vector of TOI within the imaging swath is written as

$$\mathbf{R}_{\mathbf{TOI(ECEF)}} = \mathbf{U}_{\mathbf{ECEF}}^{\mathbf{ACS}}\left[\mathbf{R}_{\mathbf{SAR(ACS)}} - \mathbf{R}_{\mathbf{ST(ACS)}}\right] \tag{A1}$$

where $\mathbf{U}_{\mathbf{ECEF}}^{\mathbf{ACS}}$ is the transformation matrix from ACS to ECEF coordinate, and its expression is given by

$$\mathbf{U}_{\mathbf{ECEF}}^{\mathbf{ACS}} = \mathbf{U}_{\mathbf{ECEF}}^{\mathbf{ECI}}\mathbf{U}_{\mathbf{ECI}}^{\mathbf{ACS}} = [U_{mn}], \quad 1 \le m \le 3, \ 1 \le n \le 3. \tag{A2}$$

Then, it is possible to express the TOIs position vector in the ECEF coordinate by

$$\begin{cases} X_{\mathrm{TOI}} = U_{11}(R_{\mathrm{ST}}\cos\theta_l - R_{\mathrm{SAR}}) \\ \qquad - U_{12}\,R_{\mathrm{ST}}\,\xi_{\mathrm{sld}}\cos\phi_a\sin\theta_l - U_{13}\,R_{\mathrm{ST}}\,\xi_{\mathrm{sld}}\sin\phi_a\sin\theta_l \\ Y_{\mathrm{TOI}} = U_{21}(R_{\mathrm{ST}}\cos\theta_l - R_{\mathrm{SAR}}) \\ \qquad - U_{22}\,R_{\mathrm{ST}}\,\xi_{\mathrm{sld}}\cos\phi_a\sin\theta_l - U_{23}\,R_{\mathrm{ST}}\,\xi_{\mathrm{sld}}\sin\phi_a\sin\theta_l \\ Z_{\mathrm{TOI}} = U_{31}(R_{\mathrm{ST}}\cos\theta_l - R_{\mathrm{SAR}}) \\ \qquad - U_{32}\,R_{\mathrm{ST}}\,\xi_{\mathrm{sld}}\cos\phi_a\sin\theta_l - U_{33}\,R_{\mathrm{ST}}\,\xi_{\mathrm{sld}}\sin\phi_a\sin\theta_l. \end{cases} \tag{A3}$$

The Earth ellipsoid imposes the condition that

$$\frac{X_{\mathrm{TOI}}^2 + Y_{\mathrm{TOI}}^2}{R_{\mathrm{eq}}^2} + \frac{Z_{\mathrm{TOI}}^2}{R_p^2} = 1. \tag{A4}$$

Upon substituting (A3) into (A4), one arrives at a quadratic equation regarding the slant range R_{ST}

$$S_1 \cdot R_{\mathrm{ST}}^2 - 2S_2 \cdot R_{\mathrm{ST}} + S_3 = 0 \tag{A5}$$

with

$$\begin{cases} S_1 = R_p^{-2}s_3^2 + R_{\mathrm{eq}}^{-2}(s_1^2 + s_2^2) \\ S_2 = -R_{\mathrm{SAR}}\left[R_{\mathrm{eq}}^{-2}(U_{11}s_1 + U_{21}s_2) + R_p^{-2}U_{31}s_3\right] \\ S_3 = R_{\mathrm{SAR}}^2\left[R_{\mathrm{eq}}^{-2}(U_{11}^2 + U_{21}^2) + R_p^{-2}U_{31}^2\right] - 1 \\ s_1 = -U_{11}\cos\theta_l + U_{12}\xi_{\mathrm{sld}}\cos\phi_a\sin\theta_l + U_{13}\xi_{\mathrm{sld}}\sin\phi_a\sin\theta_l \\ s_2 = -U_{21}\cos\theta_l + U_{22}\xi_{\mathrm{sld}}\cos\phi_a\sin\theta_l + U_{23}\xi_{\mathrm{sld}}\sin\phi_a\sin\theta_l \\ s_3 = -U_{31}\cos\theta_l + U_{32}\xi_{\mathrm{sld}}\cos\phi_a\sin\theta_l + U_{33}\xi_{\mathrm{sld}}\sin\phi_a\sin\theta_l. \end{cases} \tag{A6}$$

Solving (A5), one can determine the slant range (while discarding another solution that is meaningless)

$$R_{\mathrm{ST}} = \frac{S_2 - \sqrt{S_2^2 - S_1S_3}}{S_1}. \tag{A7}$$

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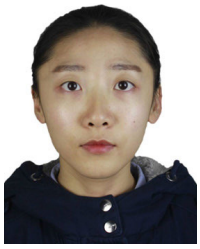
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